SDG BLOCKCHAIN ACCELERATOR

Debugging and Testing Report

In collaboration with

UNDP Bosnia and Herzegovina / University of Sarajevo) / Sala

1. Project Information

- **Project Name:** Digital Identity & Credential Verification
- **Challenge & UNDP Office:** UNDP Bosnia and Herzegovina & Challenge Owner: University of Sarajevo
- **Report Version:** V 1.0 January 2026

2. Testing Approach

(Explain how you tested your Aiken smart contracts. Include both unit tests and testnet/emulator tests.)

- **Unit Testing:** Describe test coverage, structure of Aiken unit tests, and edge cases checked.
- **Integration Testing:** Describe emulator or testnet runs.
- **Edge Cases:** Mention invalid datum/redeemer, unauthorized signatures, insufficient collateral, etc.

Example Entry:

- Unit tests written for validator logic in Aiken (`aiken test`)
- Integration tests conducted on Cardano Preview Testnet
- Edge cases:
    - Transaction with missing redeemer rejected
    - Transaction with invalid datum rejected
    - Double spend attempt blocked

3. Error Logs

(Paste relevant error logs, CLI outputs, or screenshots from emulator/testnet runs.)

Error 1:

```
 [FAIL] test_invalid_datum
Expected: validation failure
Got: transaction applied successfully
```

- **Error 2:**
  Screenshot of failed transaction on Preview Testnet (attach if available).

4. Resolved Issues

(List issues found and how they were resolved. Use a table if needed.)

Issue ID	Description	Root Cause	Resolution	Status
001	Redeemer type mismatch	Validator not checking redeemer schema	Updated validator to enforce redeemer type check	✅ Fixed
002	Script exceeded execution units	Recursive function not optimized	Refactored function to reduce CPU cost	✅ Fixed

5. Optimization Notes

(Document improvements made during debugging to enhance performance and reduce costs.)

- Script size reduced from 18KB → 12KB
- Execution units optimized:
  - CPU: 480,000 → 310,000
  - Memory: 1.5M → 900K
- Validator refactored for readability and maintainability

6. Tools and Environments Used

(List all tools, versions, and environments used in debugging and testing.)

- Aiken CLI v1.0.12 (`aiken check`, `aiken build`, `aiken test`)

- Cardano-node 8.9.0 (Preview Testnet)
- Ogmios (local chain-sync + tx monitoring)
- Kupo (for UTxO indexing)
- Local Emulator for dry-run testing

7. Remaining Issues / Next Steps

(Mention unresolved issues, pending optimizations, or planned next steps.)

Example:

- Pending improvement: validator script size still close to 16KB limit — exploring refactoring.
- Next step: run contract stress tests on Cardano Preprod Testnet.